4. In each of the following, find the value of '$a$' for which the given points are collinear.
   (i) (2, 3), (4, $a$) and (6, –3)  (ii) ($a$, 2–2$a$), (–$a$+1, 2$a$) and (–4–$a$, 6–2$a$)

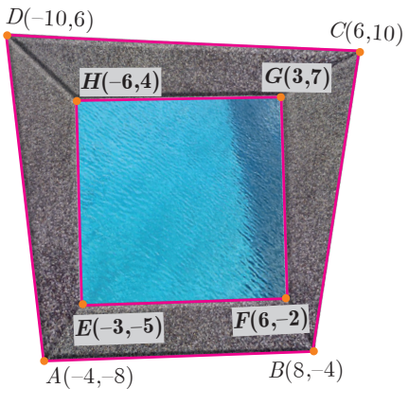
5. Find the area of the quadrilateral whose vertices are at
   (i) (–9, –2), (–8, –4), (2, 2) and (1, –3)  (ii) (–9, 0), (–8, 6), (–1, –2) and (–6, –3)

6. Find the value of $k$, if the area of a quadrilateral is 28 sq.units, whose vertices are taken in the order (–4, –2), (–3, $k$), (3, –2) and (2, 3)

7. If the points $A(-3,9)$, $B(a,b)$ and $C(4,-5)$ are collinear and if $a+b=1$, then find $a$ and $b$.

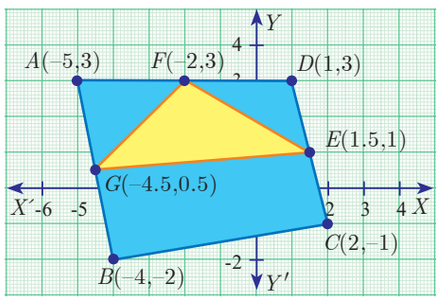
8. Let $P(11,7)$, $Q(13.5,4)$ and $R(9.5,4)$ be the mid-points of the sides $AB$, $BC$ and $AC$ respectively of $\Delta ABC$. Find the coordinates of the vertices $A$, $B$ and $C$. Hence find the area of $\Delta ABC$ and compare this with area of $\Delta PQR$.

9. In the figure, the quadrilateral swimming pool shown is surrounded by concrete patio. Find the area of the patio.




10. A triangular shaped glass with vertices at $A(-5,-4)$, $B(1,6)$ and $C(7,-4)$ has to be painted. If one bucket of paint covers 6 square feet, how many buckets of paint will be required to paint the whole glass, if only one coat of paint is applied.

11. In the figure, find the area of (i) triangle $AGF$ (ii) triangle $FED$ (iii) quadrilateral $BCEG$.




## 5.4 Inclination of a Line

The inclination of a line or the angle of inclination of a line is the angle which a straight line makes with the positive direction of $X$ axis measured in the counter-clockwise direction to the part of the line above the $X$ axis. The inclination of the line is usually denoted by $\theta$.

**Note**

- The inclination of $X$ axis and every line parallel to $X$ axis is $0°$.
- The inclination of $Y$ axis and every line parallel to $Y$ axis is $90°$.

### 5.4.1 Slope of a Straight line

While laying roads one must know how steep the road will be. Similarly, when constructing a staircase, we should consider its steepness. For the same reason, anyone travelling along a hill or a bridge, feels hard compared to travelling along a plain road.



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